
PREDICTING VA CAR CRASH SEVERITY

Harsh Gowrisetty
College of Arts and Sciences
University of Virginia
Charlottesville, VA
xxt6xz@virginia.edu

Sanjitha Prabakaran
College of Arts and Sciences
University of Virginia
Charlottesville, VA
fyw7ew@virginia.edu

Anjana Raman
School of Engineering
University of Virginia
Charlottesville, VA
yvh7bw@virginia.edu

May 2, 2025

ABSTRACT

This project analyzes Virginia car crash data to develop a predictive model that estimates accident severity (ranging from minor injuries to fatal outcomes) based on real-time conditions. Testing with both of the machine learning techniques LinearSVC and Random Forest, the system integrates diverse factors including weather, time of day, passenger count, and trip distance to accurately forecast crash severity levels (K, A, B, C, O). The goal is to provide Virginia drivers with data-driven safety insights by identifying high risk factors correlated with greater crash severity, helping them make informed decisions and reduce accident risks, initially through a model's prediction that could eventually be integrated into a mobile application for users.

1 Introduction

This project aims to address the issue of car crashes, which claim over 38,000 lives annually in the United States. In Virginia, there were 907 fatalities in 2023 . The goal is to identify lesser-known factors correlated with car accidents and develop an innovative solution that proactively helps reduce crashes. By analyzing driving circumstances, the **application** will provide drivers with informed recommendations on whether to drive based on current conditions and potentially save hundreds of lives in Virginia and tens of thousands nationally. Predictive Modeling has been leveraged for traffic accidents in previous studies such as "A Machine Learning Approach to Traffic Accident Prediction" (Yang et al., 2024). The study used various machine learning algorithms, including Light Gradient Boosting Model (LGBM), Random Forest (RF), and Neural Network (NN) to quantify the correlations of driver distraction-related crashes on road and traffic environments on rural highways. Based on the findings of this study, the team will focus on Virginia-specific driving data to refine accident prediction and better serve the Virginia community.

2 Method

The machine learning techniques employed in this study include Random Forest Classifier, Logistic Regression, LinearSVC, and a Dummy Classifier as a baseline. Prior to modeling, the dataset was preprocessed by imputing missing values, standardizing numerical features, and applying one-hot encoding to categorical variables, with high-cardinality features capped to limit dimensionality. The data was then split into training, validation, and test sets to prevent leakage. The Dummy Classifier, which assigns the most frequent crash severity (O) to every instance, establishes a baseline accuracy of 67%, reflecting that the 'O' class occurs in 67% of the cases. In contrast, the Random Forest Classifier was chosen for its capacity to capture complex, non-linear interactions among features such as weather conditions, time of day, and trip distance, while LinearSVC efficiently handles high-dimensional data and reveals the influence of different features on crash severity. Additionally, Logistic Regression will be evaluated as a simpler and more interpretable model, offering insights into feature importance through its coefficients, and serving as a useful benchmark against the more complex methods. Comparing these models makes it possible to evaluate the benefits of advanced algorithms over a trivial baseline.

3 Experiments

To evaluate model performance in predicting car crash severity, four models were tested: Dummy Classifier, Logistic Regression, Random Forest, and LinearSVC. Each model was trained on a dataset of over 200,000 Virginia crash reports, containing more than 60 features including weather, lighting, passenger count, and distance traveled.

The Dummy Classifier, which always predicts the most frequent severity class ('O' for no injury), achieved 67% accuracy to set a baseline but failed to provide meaningful predictive insights.

Logistic Regression, a linear model known for its interpretability, yielded near-perfect metrics—100% accuracy, precision, recall, and F1-score across all severity classes. However, such results strongly indicated overfitting, as perfect performance across all classes is atypical in real-world scenarios.

The Random Forest Classifier, an ensemble of decision trees, achieved high performance with an overall accuracy of 97.7%. It performed well on majority classes such as 'O' and 'C', with F1-scores close to 0.99. However, performance declined on underrepresented classes such as 'K' (fatal crashes), where recall dropped to 0.33.

LinearSVC, a support vector classifier, offered a more balanced approach. It achieved 98% overall accuracy and demonstrated realistic per-class performance. For fatal crashes (K), it attained a perfect recall of 1.00 and a precision of 0.39, indicating successful identification of all true fatal crashes with a moderate number of false positives. This model also recorded macro-average metrics of 85% precision, 97% recall, and 88% F1-score, reflecting better generalization.

Logistic Regression coefficients were used to assess feature importance. Nighttime driving, unbelted passengers, high passenger counts, and adverse weather conditions were among the most influential predictors of crash severity.

Based on these results, the next development stage involves building a mobile application to provide drivers with real-time risk scores derived from model outputs. This application would deliver pre-drive safety alerts tailored to environmental and situational conditions.

4 Results

To evaluate the models' performance in predicting car crash severity the project focused on three classifiers: Random Forest, Logistic Regression, and Linear SVM. These models were trained on preprocessed crash data from Virginia, where the target label was CRASH SEVERITY with five classes: O (no injury), C (possible injury), B (minor injury), A (serious injury), and K (fatality).

Standard metrics were used including accuracy, precision, recall, and F1-score, along with confusion matrices to assess model effectiveness across classes.

The first model, Random Forest, yielded a 98% precision, 99% accuracy, and 98% F-1 score. These are quite high metrics, and could be indicators of overfitting. The parameters for this model were : 50 estimators, 10 levels for max depth, 5 min samples before split, 3 min samples for being at a leaf node, sqrt for max features.

The second model, Logistic Regression, yielded similar results: 99% precision, 99% recall, 99% F-1 score. This model also shows indications of overfitting.

Lastly, the Linear SVC, yielded a result of 98% precision, 98% recall, and 98% F-1 score. However, the macro averages are 85% precision, 97% recall, and 88% F-1 score. These scores show that this model, although having lower averages, might be more generalizable to a new set.

This repository contains the code that yielded these results: <https://github.com/anjanaraman04/ML4VAProject.git>

5 Conclusions

The findings of this study show that machine learning models are capable of predicting car crash severity using structured historical data and relevant driving features. While Logistic Regression exhibited perfect scores suggestive of overfitting, Random Forest and LinearSVC offered better generalization to unseen data.

Among the tested models, LinearSVC presented the most practical performance—capturing all fatal crashes (Class K) while maintaining high recall and balanced metrics across classes. This behavior aligns well with safety-oriented applications, where it is preferable to err on the side of caution when predicting critical outcomes.

Insights derived from model coefficients highlighted key risk factors, such as driving at night, weather conditions, and unbelted passengers. These findings are highly actionable and can be directly used to inform driver behavior.

Future work will focus on integrating the model into a mobile application that delivers real-time crash risk assessments based on current conditions. Mock-ups of this mobile application have been created and are shown in Figures 1, 2, and 3. This application has the potential to reduce traffic fatalities by enabling drivers to make more informed decisions before entering the roadway.

6 Appendix

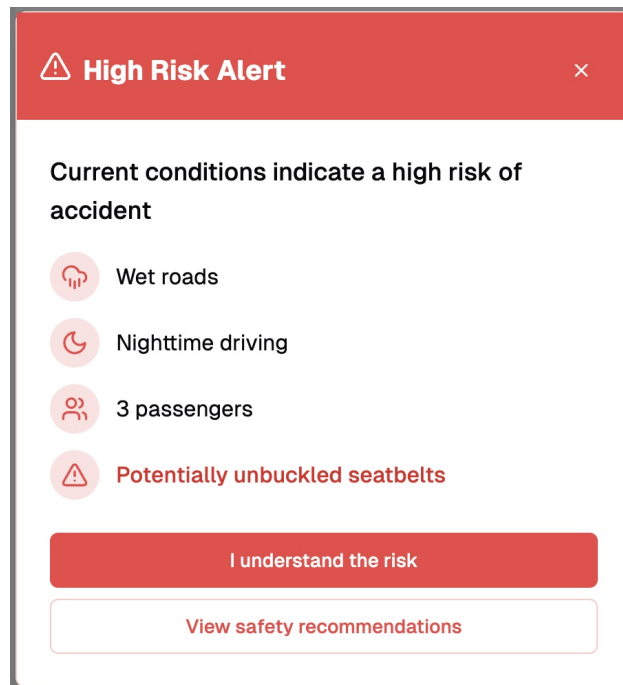


Figure 1: Application mock-up displaying crash risk alerts interface.

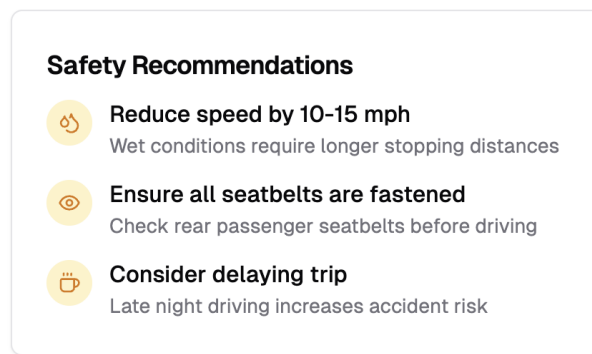


Figure 2: Application dashboard showing safety recommendations.

SafeDrive

Current Risk Assessment

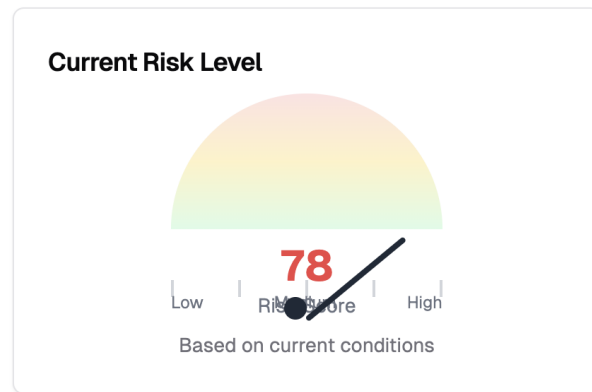
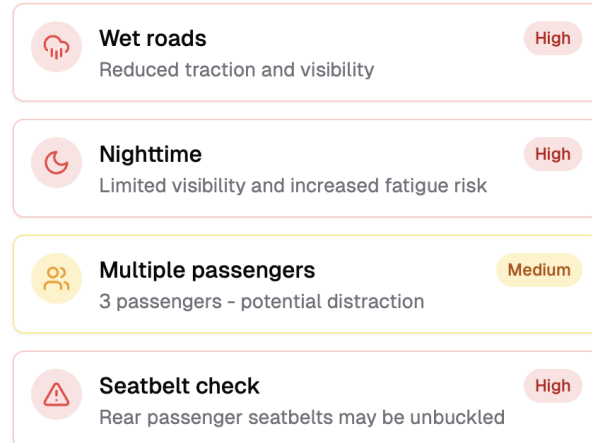
**Risk Factors**

Figure 3: Meter showing risk level and risk factors.

7 Team Contributions

Harsh Gowrisetty: Conducted data visualizations, built the preprocessing pipeline, implemented the initial Random Forest model, and helped with all writing tasks.

Sanjitha Prabakaran: Identified the dataset, refined the Random Forest implementation, ran Logistic Regression experiments, and helped with all writing tasks.

Anjana Raman: Implemented and tuned the LinearSVC model, cleaned the code repository, and helped with all writing tasks.

All team members jointly developed the ML4VA video, contributing equally to scripting, storyboarding, filming, and editing.

References

- [1] Chenxuan Yang, Jun Liu, Zihe Zhang, Emmanuel Kofi Adanu, Praveena Penmetsa, Steven Jones. A machine learning approach to understanding the road and traffic environments of crashes involving driver distraction and inattention (DDI) on rural multilane highways. In *Commonwealth Law Group Blog*, 2020. Available

- at: <https://www.sciencedirect.com/science/article/pii/S0022437524001609>. Accessed: March 7, 2025.
- [2] Commonwealth Law Group. Virginia car accident statistics: One of the most dangerous driving states. In *Elsevier*, 2024. Available at: <https://cmlawva.com/our-blog/virginia-car-accident-statistics-one-of-the-most-dangerous-driving-states/>. Accessed: March 6, 2025.
 - [3] Darrigo & Diaz Attorneys. Car accident statistics you need to know in 2021. In *Darrigo & Diaz Law Firm Blog*, 2021. Available at: <https://www.ddlawtampa.com/resources/car-accident-statistics-you-need-to-know-in-2021/>. Accessed: March 6, 2025.
 - [4] Guy Hadash, Einat Kermany, Boaz Carmeli, Ofer Lavi, George Kour, and Alon Jacovi. Estimate and replace: A novel approach to integrating deep neural networks with existing applications. *arXiv preprint arXiv:1804.09028*, 2018.
 - [5] Dan Folkes. Virginia Crash Data. In *Kaggle Datasets*, 2024. Available at: <https://www.kaggle.com/datasets/danfolkes/virginia-crash-data>. Accessed: March 6, 2025.